

STATSIG lecture 8

Spectral estimation =

how to make robust PSD estimates from data

Non-parametric (black-box) methods

(next lecture parametric methods)

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Non-parametric PSD estimation

Theoretical definition $P_x(f) = E[|X(f)|^2] = \mathcal{F}\{r_{xx}(\tau)\}$

In practice (only given a single signal realization) neither of these definitions can be used since the true autocorrelation function is not known and $E[\]$ requires probability distribution

Task Make a robust estimate, $\widehat{P}_x(f)$, based on N signal samples

Desired properties of estimator (details on blackboard)

Unbiased or asymptotically unbiased

Consistent

Small MSE

Low computational complexity

Periodogram (naive approach)

$\hat{P}_x(f) = \frac{1}{N} |X_N(f)|^2$ where $X_N(f)$ is the N-point DFT/FFT of the signal $x(n)$ of length N

Biased, but asymptotically unbiased

Not consistent

Frequency resolution $(f_2 - f_1)_{min} = 0.89 \frac{f_s}{N}$

Computational complexity $O(N \log N)$

Bartlett method (averaging of sub-periodograms)

Partition the N signal samples into K blocks of length L, $K = N/L$, take periodogram of each of the length L blocks and calculate the average

$\hat{P}_x(f) = \frac{1}{K} \sum_{i=1}^K \hat{P}_x^{(i)}(f)$ where $\hat{P}_x^{(i)}(f) = \frac{1}{L} |X_L^{(i)}(f)|^2$ (periodogram for block i)

Biased, but asymptotically unbiased as periodogram

Consistent

Frequency resolution $(f_2 - f_1)_{min} = 0.89 \frac{f_s}{L}$ (worse than Periodogram)

Computational complexity $O(N \log N)$

Welch's method (improvement of Bartlett by adding flexibility)

Averaging of periodograms for sub-blocks as Bartlett method, possibility of overlapping blocks giving more blocks to average over, possibility to use smoothing window for blocks

Windowing
$$x_{mod,L}^{(i)}(n) = x^{(i)}_L(n) \cdot w_L(n)$$

Estimate for block (i)
$$\hat{P}_x^{(i)}(f) = \frac{1}{U} \frac{1}{L} \left| X_{mod,L}^{(i)}(f) \right|^2 \quad \text{where} \quad U = \frac{1}{L} \sum_{n=0}^{L-1} w^2(n)$$

Averaging
$$\hat{P}_x(f) = \frac{1}{K} \sum_{i=1}^K \hat{P}_x^{(i)}(f)$$

Biased, but asymptotically unbiased depending of window function

Consistent

Frequency resolution $(f_2 - f_1)_{min} = 1.28 \frac{f_s}{L}$ (Hamming window)

Computational complexity $O(N \log N)$

Blackman-Tukey method

Previous methods has used $P_x(f) = E[|X(f)|^2]$
Blackman-Tukey uses $P_x(f) = \mathcal{F}\{r_{xx}(\tau)\}$

Estimate correlation function as usually $\hat{r}_{xx}(\tau) = \frac{1}{N} \sum_{n=0}^{N-1} x(n)x(n+\tau), \quad |\tau| \leq N-1$

Use only inner (most reliable) part of correlation estimate by multiplying with a window, $w(n)$,
spanning from $n = -M$ to $N = M$
As a rule of thumb $M/N = 10\text{-}20\%$

$$\hat{P}_x(f) = DFT_{2M+1}\{\hat{r}_{xx}(\tau)w_{2M+1}(\tau)\}$$

Biased, but asymptotically unbiased depending of window function
Consistent (variance depends on window and M/N)

Frequency resolution $(f_2 - f_1)_{min} = 1.28 \frac{f_s}{2M}$ (Hamming window)

Computational complexity $O(N \log N)$